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Early menopause as an independent risk factor for postoperative atrial fibrillation in women undergoing coronary artery bypass grafting: a retrospective cohort study

Haokai Qin^{1,2†}, Jiyuan Luo^{1†}, Yunxiao Yang³, Xunxun Feng^{3,4}, Xiubin Yang^{1*}, Kun Hua^{1*} and Mingyang Zhou^{1*}

Abstract

Background Early menopause is considered directly related to the occurrence of isolated atrial fibrillation. In this study, we aim to explore the relationship between early menopause and the incidence of postoperative atrial fibrillation (POAF) occurring following coronary artery bypass grafting (CABG).

Method This retrospective cohort study included patients who underwent CABG at Beijing Anzhen Hospital from January 2020 to December 2023. Patients were divided into two cohorts: early menopausal individuals (age < 45) and non-early menopausal individuals (age ≥ 45). Baseline characteristic disparities were adjusted via including all baseline variables with an standardized mean difference (SMD) > 0.1 as covariates in the inverse probability of treatment weighting (IPTW) analysis. The correlation between menopausal age and the occurrence of POAF was analyzed utilizing logistic regression and restricted cubic spline (RCS).

Results Among the 1825 female patients who underwent CABG in this study, 537 developed POAF. After IPTW analysis to control for baseline variables, it was found that the incidence of POAF in the early menopause group was significantly higher than in the non-early menopause group (42.1% vs. 28.2%). Logistic regression analysis showed that early menopause was associated with an increased risk of POAF (adjusted OR: 1.76, 95% CI: 1.30–2.38, $p < 0.001$). RCS analysis also confirmed our hypothesis, showing that early menopause is associated with an increased risk of POAF.

Conclusion Female CABG patients with early menopause have a higher incidence of POAF compared to those without early menopause. Additionally, the younger the age at menopause, the higher the risk of developing POAF.

[†]Haokai Qin and Jiyuan Luo contributed equally to this work.

*Correspondence:

Xiubin Yang
xiubinyang@hotmail.com
Kun Hua
huakun0310@126.com
Mingyang Zhou
az6885@sina.com

Full list of author information is available at the end of the article



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These findings suggest that menopausal age should be considered in preoperative risk stratification, and early menopausal women may benefit from targeted monitoring or preventive strategies to reduce POAF risk.

Keywords Early menopause, Coronary artery bypass grafting, Postoperative atrial fibrillation, Female health

Introduction

Cardiovascular diseases remain the leading cause of death globally, with postoperative atrial fibrillation (POAF) being the most common arrhythmia following cardiac surgery. According to the World Heart Report 2023, cardiovascular diseases caused 20.5 million deaths in 2021, accounting for one-third of global deaths.

Atrial fibrillation (AF) itself exhibits significant sex-based disparities in both its epidemiology and clinical outcomes. Although women generally have a lower incidence of AF compared to men, they appear to be at higher risk for arrhythmia-related complications such as stroke, heart failure, and death [1, 2]. Moreover, there is evidence specifically links menopause to increased AF risk: women with a history of early menopause face an elevated risk of AF compared to those without, with younger age at menopause correlating with higher AF risk [3].

Cardiac surgery is a common and essential treatment for cardiovascular disorders, yet it carries a significant risk of POAF, the most frequent postoperative complication with an incidence of 20–40% [2, 4]. It is defined as new-onset atrial fibrillation or flutter lasting at least one hour after surgery [5]. POAF significantly increases the incidence of major adverse cardiovascular events, including stroke, heart failure, and clinical death, while also prolonging hospital stays and increasing healthcare costs [2, 6–8]. However, the relationship between the risk of POAF and age at menopause remains unclear. The established increased cardiovascular risk and AF susceptibility associated with menopause, particularly early menopause, underscores the critical need to investigate its impact specifically on POAF risk. It's true that there are currently some effective treatments for postoperative atrial fibrillation following cardiac surgery [2], approximately 20% of cases still cannot be completely resolved, leaving the risk unabated. Therefore, preventing the occurrence of POAF and investigating the role of early menopause is of paramount importance.

Growing evidence suggests that menopause significantly increases the risk of cardiovascular diseases in women. Menopause is defined as the permanent cessation of menstruation resulting from reduced ovarian hormone production [9]. The usual onset of menopause for most women occurs between the ages of 49 and 52 [10, 11]. Early menopause, defined as menopause before age 45, affects about 10% of women [12–14]. According to the 2019 ACC/AHA Cardiovascular Disease Primary Prevention Guidelines, postmenopausal women face a

significantly increased risk of cardiovascular diseases due to the decline in estrogen levels. A meta-analysis of 15 observational studies from five countries showed that early menopause (age < 45) significantly increase the risk of cardiovascular diseases [15]. Critically, these associations persist even after adjusting for chronological age, suggesting that ovarian aging may exert cardiovascular effects through pathways distinct from somatic aging. The differential findings highlighted above emphasize the urgent need to identify and understand the risk of POAF in women with varying menopausal histories. This knowledge is paramount for developing effective POAF prevention and treatment strategies in this vulnerable population.

Although early menopause has been associated with an increased risk of cardiovascular diseases, its effect on the incidence of POAF following coronary artery bypass grafting (CABG) remains largely unknown. To address this gap, our study is the first to explore the relationship between early menopause and POAF risk, offering valuable insight for perioperative risk assessment in female patients undergoing cardiac surgery.

Subjects and methods

Establishing the study populations

This study encompassed 2,658 female patients who underwent solitary CABG surgery at Beijing Anzhen Hospital from January 2020 to December 2023. The exclusion criteria included: premenopausal status, inability to specify age of menopause, absence of endpoint data, a history of atrial fibrillation (included the use of antiarrhythmic drugs preoperatively), underwent emergency CABG surgery and concurrent surgical operations with CABG. Consequently, 833 patients were omitted from the study. A total of 1825 patients were incorporated into the analysis. Patients were categorized into two groups based on menopause age: those with menopause occurring at age ≥ 45 years were designated as the non-early menopause group ($n = 1607$), while those with menopause at age < 45 years were designated as the early menopause group ($n = 218$) (Fig. 1). The Ethics Committee of our hospital approved this study (The Ethics Committee for Clinical Research of Beijing Anzhen Hospital, approved number: 2025033x), and all patients gave written informed consent.

Data collection and patient outcomes

Patient data were obtained from the electronic medical record system, encompassing demographic details,

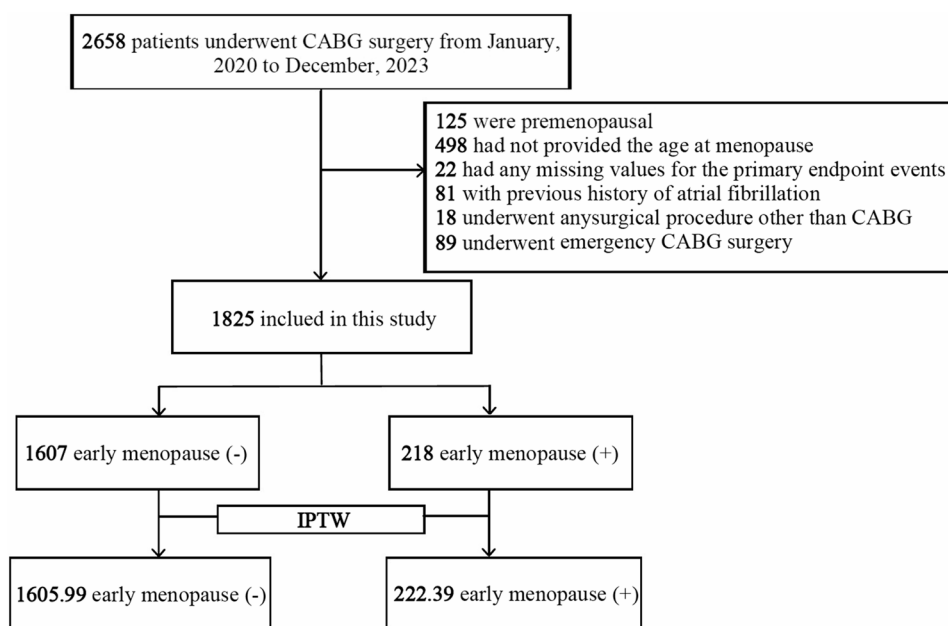


Fig. 1 Study flowchart

comorbidities, hospitalization outcomes, medications, preoperative laboratory tests, echocardiography, and continuous electrocardiogram (ECG) monitoring data. In addition, information on the age at menopause for female patients was collected from patient-reports. We define menopause as the absence of menstruation for more than one year [16]. The principal outcome was the incidence of POAF. POAF is defined as postoperative AF lasting longer than 1 h and/or requiring treatment [5]. All patients were subjected to continuous cardiac monitoring for postoperative atrial fibrillation during their ICU stay and for the initial three days following their transfer to the general ward.

Statistical analysis

The normality of continuous variables was tested using the Kolmogorov-Smirnov test. For quantitative data, variables with a normal distribution were described using means $\pm$ standard deviation, and group differences were analyzed using the Student's *t*-test. Variables with a skewed distribution were defined using the median and interquartile range (25th–75th percentiles), and group comparisons were performed using the Mann-Whitney *U* test. Differences were analyzed using the Chi-square (χ^2) test for qualitative data. Data were analyzed using inverse probability of treatment weighting (IPTW) to control for differences in baseline characteristics between the early menopause group and the non-early menopause group. The weight of IPTW were obtained from propensity score matching and computed using the “ipw” package in R software. The standardized mean difference (SMD) values for all variables were calculated,

and variables with an SMD > 0.1 (age, CK-MB, E/A, UA, operation time, hypertension) were included as covariates in the propensity score model for PSM. SMD < 0.1 were used as the criterion for non-significant differences between groups. The restricted cubic spline (RCS) model, featuring four nodes at the 5th, 35th, 65th, and 95th percentiles, was employed to examine the nonlinear dose-response relationship between menopausal age and POAF, while accounting for significant confounding variables associated with POAF. Univariate and multivariate logistic regression models were utilized to assess the independent relationship between menopausal age and the progressively rising menopausal age with the incidence of significant outcomes, represented as odds ratios (OR) and 95% confidence intervals (CI). In constructing the multivariate regression models, confounding variables were selected based on established clinical relevance and evidence from previous studies reporting risk factors associated with POAF and cardiovascular outcomes [17–20]. Variance inflation factor (VIF) values were used to assess multicollinearity among covariates. The Hosmer-Lemeshow (H-L) test was used to evaluate the goodness-of-fit of the multivariate logistic regression model. Statistical analyses were performed by R software (version 4.4.1). The statistical significance level was set at two-tailed $p < 0.05$.

Results

The baseline characteristics

This study included 1,825 female patients who underwent isolated CABG, with 218 early menopause patients and 1,607 non-early menopause patients. Notable differences

in age, CK-MB, E/A, UA, operation time and hypertension (SMD > 0.1) were seen between the early menopause group and the non-early menopause group regarding baseline patient data. After implementing IPTW, the SMD for all variables except age were < 0.1, proving that the patients' baseline data were balanced (Table 1; Fig. 2).

The relationship between early menopause and the occurrence of POAF

We compared the probability of POAF occurrence between the early menopause group and the non-early menopause group. We found a significant difference in POAF incidence after adjusting for the IPTW (42.1% vs. 28.2%, $p < 0.001$). The probability of POAF occurrence in early menopause female CABG patients was significantly higher than in non-early menopause female CABG patients. Through univariate regression analyses, early menopause was identified as independently associated with the occurrence of POAF after CABG surgery (OR: 1.85, 95% CI: 1.38–2.46, $p < 0.001$). After adjusting for the covariates again, the correlation still exists (adjusted OR: 1.76, 95% CI: 1.30–2.38, $p < 0.001$) (Table 2). The VIF values of the covariates are all less than 5, indicating that

there is no collinearity relationship among the selected covariates (sFigure1, sTable1). The p-value of the H-L test is greater than 0.05, indicating that there is no significant difference between the predicted probabilities and the actual probabilities of the multivariate logistic regression model, and the model fits well (sFigure2).

To further elucidate the relationship between early menopause and postoperative atrial fibrillation, an RCS analysis was conducted. The results indicated that after the menopausal age dropped below 49 years, the risk of POAF increased with decreasing menopausal age, and this association remained significant after adjusting for confounding factors (Fig. 3).

Subgroup analysis

We conducted a subgroup analysis of all patients based on age (using 70 years as the cutoff), history of hypertension, history of hyperlipidemia, BMI, number of grafts (using 3 grafts as the cutoff), and surgery duration (using 4 h as the cutoff). We observed that early menopause was significantly associated with an increased risk of POAF in the following groups: Women aged over 70 years (OR: 1.95, 95% CI: 1.03–3.72, $p = 0.042$); Patients with

Table 1 Baseline characteristics between early menopause (+) group and early menopause (-) group

	Unmatched			IPTW		
	Early menopause (-)	Early menopause (+)	p value	Early menopause (-)	Early menopause (+)	p value
n	1607	218		1605.99	222.39	
Age, years	65.38 ± 6.60	62.88 ± 8.53	< 0.001*	65.11 ± 6.66	62.88 ± 8.53	0.111
Menopausal age, years	49.73 (2.64)	40.93 (2.76)	< 0.001*	49.72 (2.64)	40.90 (2.84)	< 0.001*
BMI, kg/m²	25.44 ± 3.34	25.58 ± 3.36	0.56	25.45 ± 3.36	25.54 ± 3.11	0.693
Hypertension (%)	1126 (70.1)	142 (65.1)	0.16	1115.8 (69.5)	153.5 (69.0)	0.898
Diabetes (%)	732 (45.6)	99 (45.4)	1	730.7 (45.5)	1.1 (0.5)	0.318
COPD (%)	21 (1.3)	1 (0.5)	0.456	20.6 (1.3)	1.0 (0.5)	0.382
Hyperlipidemia (%)	945 (58.8)	126 (57.8)	0.834	945.9 (58.9)	134.2 (60.3)	0.710
CKD (%)	28 (1.7)	4 (1.8)	1	28.3 (1.8)	4.7 (2.1)	0.771
Stroke history (%)	240 (14.9)	31 (14.2)	0.86	237.8 (14.8)	35.7 (16.1)	0.668
Preoperative laboratory data						
TG, mmol/L	1.47 [1.14, 2.00]	1.61 [1.11, 2.18]	0.104	1.47 [1.14, 2.01]	1.54 [1.07, 2.11]	0.668
TC, mmol/L	4.24 ± 1.13	4.23 ± 1.15	0.952	4.24 ± 1.13	4.22 ± 1.15	0.804
Cr, μmol/L	57.80 [49.30, 67.95]	59.05 [49.10, 67.70]	0.435	57.80 [49.30, 67.94]	59.22 [49.40, 68.21]	0.234
UA, μmol/L	288.80 [240.55, 350.15]	294.65 [243.55, 357.80]	0.213	289.80 [241.52, 351.27]	279.36 [239.37, 345.58]	0.561
CK-MB, ng/ml	3.00 [1.60, 6.20]	2.15 [1.40, 4.30]	< 0.001	3.00 [1.60, 6.10]	2.55 [1.40, 5.25]	0.211
TnI, pg/ml	0.14 [0.05, 0.44]	0.16 [0.05, 0.54]	0.343	0.14 [0.05, 0.44]	0.17 [0.06, 0.63]	0.071
BNP, pg/ml	210.00 [88.00, 336.00]	228.00 [98.00, 353.75]	0.449	209.00 [86.00, 334.06]	241.45 [113.79, 362.72]	0.137
Preoperative echocardiographic data						
LAD, mm	35.59 ± 3.93	35.69 ± 3.50	0.699	35.57 ± 3.94	35.68 ± 3.70	0.712
LVEF, %	61.37 ± 6.98	61.10 ± 6.74	0.59	61.35 ± 7.00	61.24 ± 6.82	0.847
E/A ratio	0.78 ± 0.26	0.82 ± 0.24	0.024	0.78 ± 0.26	0.78 ± 0.22	0.919
LVEDd, mm	29.99 ± 4.27	30.28 ± 4.24	0.344	30.00 ± 4.28	29.89 ± 4.15	0.750
CABG data						
LIMA usage, %	918 (57.1)	126 (57.8)	0.908	925.9 (57.7)	115.4 (51.7)	0.136
On-pump CABG (%)	118 (7.3)	14 (6.4)	0.724	115.9 (7.2)	18.5 (8.3)	0.650
Graft number, n	2.66 ± 0.94	2.63 ± 0.86	0.673	2.65 ± 0.95	2.69 ± 0.84	0.585
Operation time, h	4.12 ± 1.30	3.99 ± 1.06	0.163	4.10 ± 1.28	4.14 ± 1.19	0.752

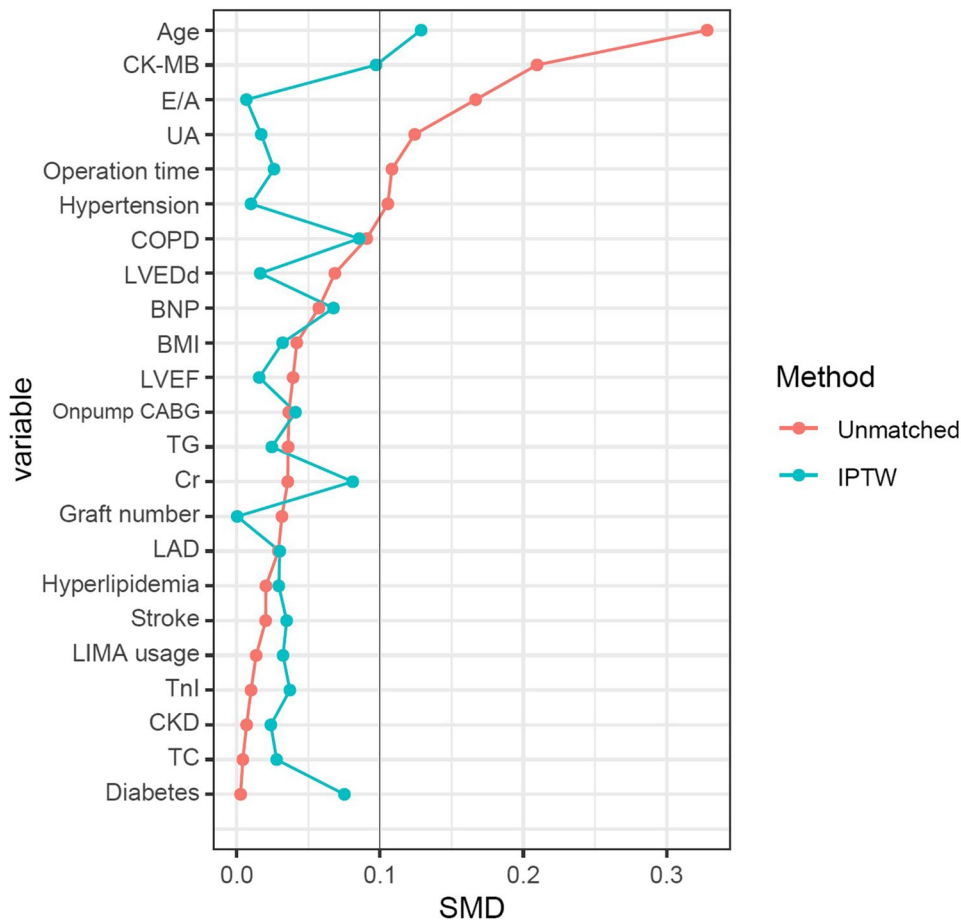


Fig. 2 SMD values for all variables before IPTW (red) and after IPTW adjustment (blue)

Table 2 Incidence and risk of POAF associated with early menopause after IPTW analysis

POAF	Early menopause (-)		Early menopaus (+)		Odds Ratio (95% confidence interval)		
	Event	Incidence rate	Event	Incidence rate	Model 1 ^a	Model 2 ^b	Model 3 ^c
No					1 (reference)	1 (reference)	1 (reference)
Yes	453.0 (1605.99)	28.2%	93.5 (222.39)	42.1%	1.85 [1.38, 2.46]	1.77 [1.31, 2.38]	1.76 [1.30, 2.38]
P value					< 0.001	< 0.001	< 0.001

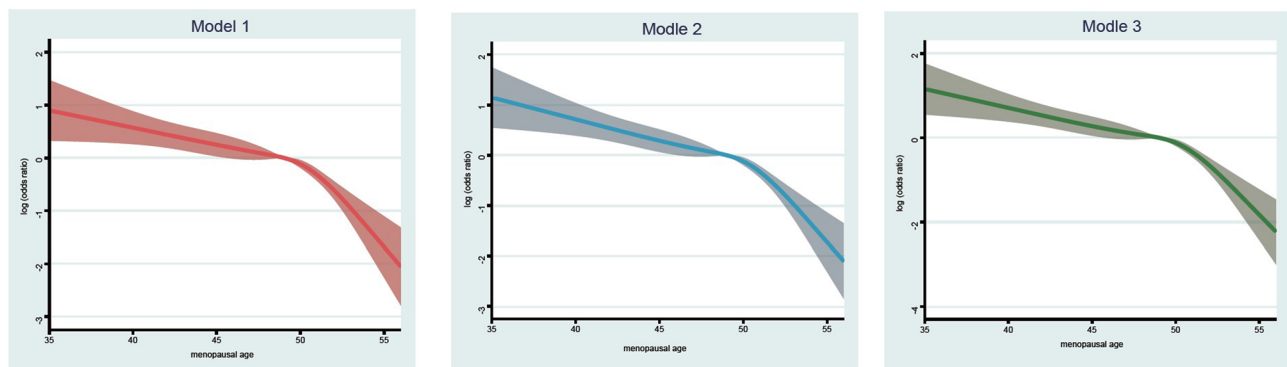
^aModel 1 was not adjusted
^bModel 2 was adjusted for age, hypertension and diabetes
^cModel 3 was adjusted for age, hypertension, diabetes, BMI, LVEF, operation time and TnI

a history of hypertension (OR: 1.67, 95% CI: 1.16–2.39, $p=0.005$); Patients with a history of hyperlipidemia (OR: 1.50, 95% CI: 1.02–2.21, $p=0.042$); Patients with BMI ≤ 24 (OR: 1.86, 95% CI: 1.07–3.23, $p=0.029$); Patients with ≥ 3 grafts (OR: 1.55, 95% CI: 1.04–2.29, $p=0.031$); Patients with surgery duration exceeding 4 h (OR: 2.66, 95% CI: 1.57–4.49, $p<0.001$). Furthermore, interaction analysis revealed that the effect of surgery duration exceeding 4 h on POAF was influenced by early menopause (p for interaction = 0.006) (Fig. 4). Subgroup analysis also showed that patients aged over 70 years were still affected by early menopause. Therefore, we further assessed whether the time span from the last menopause to surgery played

a key role in this association. When we replaced menopausal age with the time interval from last menopause to surgery in the multivariable logistic regression model, the interval was significantly associated with POAF risk (OR: 1.08, 95% CI: [1.07, 1.10], $p<0.001$), indicating that a longer postmenopausal period may contribute to higher POAF incidence.

Discussion

This study revealed that female CABG patients experiencing early menopause had a significantly higher incidence of POAF compared to those without early menopause. After IPTW adjustment for baseline



Model 1 was not adjusted.

Model 2 was adjusted for age, hypertension and diabetes.

Model 3 was adjusted for age, hypertension, diabetes, BMI, LVEF, operation time and TnI.

Fig. 3 RCS analysis of the association between menopausal age and POAF

Variable	Count	Percent		OR (95% CI)	P value	P for interaction
Overall	1825	100		1.43 (1.06 to 1.92)	0.019	
Age						0.32
> 70	400	21.9		1.95 (1.03 to 3.72)	0.042	
<= 70	1425	78.1		1.35 (0.96 to 1.90)	0.086	
Hypertension						0.154
No	557	30.5		1.04 (0.61 to 1.78)	0.874	
Yes	1268	69.5		1.67 (1.16 to 2.39)	0.005	
Hyperlipidemia						0.696
No	754	41.3		1.33 (0.84 to 2.10)	0.222	
Yes	1071	58.7		1.50 (1.02 to 2.21)	0.042	
BMI						0.253
> 24	1231	67.5		1.27 (0.89 to 1.80)	0.188	
<= 24	594	32.5		1.86 (1.07 to 3.23)	0.029	
Graft number						0.585
>= 3	1045	57.3		1.55 (1.04 to 2.29)	0.031	
< 3	780	42.7		1.31 (0.83 to 2.06)	0.247	
Operation time						0.006
> 4 hours	660	36.2		2.66 (1.57 to 4.49)	<0.001	
<= 4 hours	1165	63.8		1.08 (0.74 to 1.55)	0.7	

Fig. 4 Subgroup analysis results

confounders, the association between early menopause and POAF persisted. Furthermore, a dose-response relationship was observed, with younger menopausal age correlating with increased POAF likelihood.

CABG is one of the major surgical treatments for coronary artery disease. However, it is prone to postoperative complications, with POAF being the most common. Numerous studies have shown that POAF is an independent predictor of poor postoperative outcomes. It can

cause hemodynamic instability, leading to an increased risk of ischemic stroke, heart failure, respiratory failure, cardiac arrest, and other complications, significantly impairing the prognosis after CABG surgery [21]. Estrogen plays a crucial protective role in women's cardiovascular health. Before menopause, higher estrogen levels contribute to favorable lipid profiles, enhanced endothelial function, and reduced vascular inflammation, collectively lowering the risk of coronary artery disease (CAD).

However, the onset of menopause leads to a significant decline in estrogen levels, resulting in adverse changes such as increased low-density lipoprotein cholesterol, decreased high-density lipoprotein cholesterol, and heightened arterial stiffness [22, 23]. These alterations elevate the risk of developing CAD. Notably, women who experience early menopause face an even greater risk, with studies indicating a 15%–50% increased likelihood of CAD compared to those who undergo menopause at the average age. This heightened risk is attributed to the prolonged duration without estrogen's protective effects [15, 24, 25].

Beyond its influence on CAD, estrogen deficiency has been implicated in the development of postoperative atrial fibrillation. The observed association may be partly mediated by estrogen depletion. While chronological age is a recognized major risk factor for POAF, reproductive aging—particularly early menopause—may independently influence arrhythmia risk, even after adjusting for age [14]. The abrupt decline in estrogen levels preceding the expected age of menopause may exacerbate pro-arrhythmic substrate formation, including atrial fibrosis, autonomic imbalance, and heightened systemic inflammation, independently amplifying susceptibility to POAF. Additionally, women with early menopause have a significantly higher risk of developing cardiovascular diseases, with each one-year delay in menopause reducing the risk of cardiovascular mortality by 2% [26]. Furthermore, some literature reports that early menopause is a triggering factor for lone atrial fibrillation, with a positive correlation between early menopause and its occurrence. A study by Jean Shin et al., which included 1,401,175 postmenopausal women, found that the risk of atrial fibrillation was significantly higher in women with early menopause, and as the age of menopause decreases, the risk increases accordingly [3].

Since cardiomyocytes express estrogen and androgen receptors, sex hormones can influence the occurrence of atrial fibrillation by affecting ion channel expression [27, 28]. Studies by Liang et al. have found that estrogen reduces atrial two-pore potassium ion channels, leading to prolonged atrial action potential duration and a proarrhythmic effect. This has been demonstrated in patients with lone atrial fibrillation carrying KCNK3/TWIK-1 loss-of-function mutations. Additionally, estrogen can enhance atrial fibrillation-triggering activity by increasing L-type calcium channels [29], sodium-calcium exchanger activity, and myocardial ryanodine receptor 2 leakage, contributing to the onset, progression, and maintenance of atrial fibrillation.

Through the influence of inflammation, early menopause may also raise the risk of POAF by encouraging atrial fibrosis and electrical remodeling. The atrium's structural matrix and electrophysiology can be changed

by mediators of the inflammatory response, increasing the risk of atrial fibrillation [30, 31]. Additionally, inflammation affects connexins and calcium homeostasis, which are connected to abnormal atrial conduction and AF triggers. Atria structural remodeling can be facilitated by inflammatory pathways, which also cause myolysis, cardiomyocyte death, and the activation of fibrotic pathways through fibroblasts, transforming growth factor- β , and matrix metalloproteinases [32]. Given this inflammatory interplay, preoperative assessment of systemic inflammation markers (e.g. IL-6, CRP) in early menopausal women may help identify high-risk subgroups who could benefit from targeted anti-inflammatory prophylaxis, such as low-dose colchicine - an approach recently shown to reduce POAF incidence in pilot trials [33].

Subgroup analysis in our study revealed that the time span from last menopause to surgery significantly associated with POAF risk, suggesting that a longer post-menopausal period may contribute to higher POAF incidence. This finding implies that late menopause could potentially confer a protective effect. However, given that advanced age is itself a strong independent risk factor for POAF, it remains uncertain whether the increased risk stems from a longer estrogen-deficient state or simply reflects the aging process. This distinction warrants further exploration in future studies.

However, although the aforementioned mechanisms provide a theoretical basis for the potential link between early menopause and POAF, the exact mechanism underlying this relationship remains unclear. The retrospective design precludes causal inference, residual confounding (e.g. unmeasured genetic, socioeconomic, or lifestyle factors) may persist despite IPTW adjustment. In particular, although IPTW effectively balances measured baseline variables, unmeasured or unknown confounders could still influence the observed associations. Second, menopausal age was obtained from patients-reports, which may lack precision compared to medical records and could introduce misclassification bias. Additionally, our study was conducted at a single center in China, which may limit the generalizability of the findings to other populations with different ethnic, geographic, and healthcare characteristics. Most critically, while estrogen replacement therapy (ERT) hypothetically could mitigate POAF risk in early menopause, our data do not support clinical ERT recommendations without randomized trial evidence. Therefore, future research should further investigate and explore the association between early menopause and POAF, as well as potential interventions, such as estrogen replacement therapy or lifestyle modifications, to reduce the risk of POAF.

We believe that early menopause is an independent risk factor for POAF, potentially linked to decreased estrogen

levels and enhanced inflammatory responses. For female patients with early menopause undergoing CABG surgery, it is crucial to emphasize risk assessment for POAF and implement personalized preventive and therapeutic strategies to reduce its incidence.

Conclusion

We found that female patients with early menopause undergoing CABG surgery have a higher incidence of POAF compared to those without early menopause. Overall, the earlier the age of menopause, the higher the proportion of female CABG patients who develop POAF. Therefore, closely monitoring the menstrual history of female patients before CABG surgery, proactively monitoring and preventing POAF in female patients with early menopause and actively intervening once it is detected may help optimize the prognosis of these patients.

Limitation

Beyond the limitations acknowledged in the discussion section, additional limitations exist as follows:

First, the retrospective design constrained our inclusion of potentially significant variables, such as postoperative vasoactive drug administration and intraoperative factors, which may yield novel insights into POAF correlations. More critically, this design inherently precludes causal inference regarding the relationship between early menopause and POAF. Second, although the proposed mechanisms provide a theoretical basis for the link, the exact biological pathway underlying the relationship between early menopause and POAF remains unclear. Third, our study had a limited sample size and primarily focused on in-hospital POAF, lacking long-term follow-up data to assess the potential association between early menopause and the occurrence of atrial fibrillation after discharge. Therefore, larger-scale prospective studies with long-term follow-up are needed to validate our findings.

Abbreviations

POAF	Postoperative atrial fibrillation
CABG	Coronary artery bypass grafting
IPTW	Inverse probability of treatment weighting
RCS	Restricted cubic spline
ECG	Electrocardiogram
SMD	Standardized mean difference
OR	Odds ratios
CI	Confidence intervals
BMI	Body mass index
COPD	Chronic obstructive pulmonary disease
CKD	Chronic kidney disease
TG	Triglyceride
TC	Total cholesterol
Cr	Creatinine
UA	Uric acid
CK-MB	Creatine kinase MB
TnI	Troponin I
BNP	Brain natriuretic peptide
LAD	Left atrial diameter

LVEF	Left ventricular ejection fractions
E/A	Ratio early to late diastolic transmitral flow velocity
LVEDd	Left ventricular end-diastolic dimension
LIMA	Left internal mammary artery
VIF	Variance inflation factor
H-L test	Hosmer-Lemeshow test

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12905-025-03941-7>.

Supplementary Material 1: sFigure1. VIF plot. sFigure2. The result of Hosmer-Lemeshow test for logistic regression model. sTable 1. VIF of variables in logistic regression model

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Not applicable.

Authors' contributions

M.Z. conceptualized the study and reviewed the manuscript, while K.H. and X.Y. refined the manuscript and provided guidance on the study design. H.Q. analyzed the data. H.Q. and J.L. wrote the manuscript. Y.Y. provided the idea, while X.F. revised and refined the manuscript. All authors contributed to editorial changes in the manuscript.

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Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to the nature of this research, participants of this study did not agree for their data to be shared publicly but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The Ethics Committee of our hospital approved this study (The Ethics Committee for Clinical Research of Beijing Anzhen Hospital, approved number: 2025033x), and all patients gave written informed consent.

Consent for publication

Not applicable.

Competing Interests

The authors declare no competing interests.

Author details

¹Department of Cardiovascular Surgery, Beijing Anzhen Hospital, Capital Medical University, Beijing 100029, China

²Chinese Institutes for Medical Research, Beijing 100069, China

³Department of Cardiology, Beijing Anzhen Hospital, Capital Medical University, Beijing 100029, China

⁴Department of Medicine, Division of Cardiology, University of California, Los Angeles, CA, USA

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